Autonomous Campus CCTV Waste Surveillance Pipeline

8-Stage Computer Vision, Stratified Dynamic Auditing & BVG Dispatch Specification

Institution: Chhatrapati Shahu Ji Maharaj University (CSJMU), Kanpur • **Division:** Smart Campus Urban Planning Cell
Architecture Build: v2.4 (Production Institutional Release) • **Surveillance Area:** 264-Acre Campus • **Inference Model:** YOLOv11-TACO (640×640)

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1. Executive Summary & Operational Engineering Principles

Monitoring sanitation compliance across CSJMU's 264-acre campus requires high-frequency surveillance without imposing unrealistic network overhead or staffing costs. Standard campus CCTV systems (CP PLUS, Hikvision, Dahua) cannot sustain continuous 4K @ 30 FPS video streaming across campus core switches without severe bandwidth saturation. Furthermore, standard YOLO models downsample inputs to **640×640 tensors**, making full-frame 4K streaming wasteful in both compute cycles and network capacity.

- **1. Ephemeral Sub-Stream / Snapshot Ingestion:** Instead of opening persistent 24/7 RTSP sessions, the pipeline queries camera HTTP snapshot APIs (/ISAPI/Streaming/channels/101/picture) or pulls from RTSP sub-streams (Channel 102). Network bandwidth consumption drops by over 99.8%.
- **2. Stratified Dynamic Random Auditing:** Fixed sweep times (e.g. 6 AM, 12 PM) create the 'anticipation effect,' where cleaning teams only tidy up minutes before scheduled sweeps. The pipeline implements a stratified pseudo-random scheduler that computes unpredictable audit timestamps inside 4 operational shift quarters (06:00 to 22:00) with a minimum 90-minute spacing buffer.
- **3. Centralized Datacenter Inference:** Because audits occur only 4 times daily across 10–30 cameras, central server inference processes all camera frames in **under 2.5 seconds total**, eliminating expensive and failure-prone outdoor edge hardware (Jetson) on camera poles.
- **4. Closed-Loop Municipal Action:** Optical detections feed directly into the CSJMU Ward Authority Matrix and BVG Sanitation Beat crews, completing work order generation, SLA tracking, and before-vs-after verification.

2. The 8-Stage CCTV Pipeline Architecture

The pipeline executes an autonomous 8-step sequence that transforms raw campus camera optics into verified sanitation ticket closures:

Stage	Pipeline Stage Name	Technical Process & Mechanism	Output / Artifact
Stage 1	**CCTV Data Collection**	Stratified random trigger activates sweep across camera registry across 4 shift windows.	Triggered Camera Manifest
Stage 2	**Data Ingestion & Storage**	HTTP GET ISAPI snapshot capture (~180 KB), letterbox resize, face/license privacy blur.	Clean 640x640 Input Buffer
Stage 3	**AI Computer Vision**	YOLOv11-TACO inference (6 classes), confidence gate (≥0.70), volumetric & weight estimation.	Bounding Box, Class, Weight (kg)
Stage 4	**Geotagging & GIS**	Camera ID lookup against campus spatial database; tags 2D floor coordinates and WGS84 GPS.	Geolocated Telemetry Package
Stage 5	**LLM Structured Parsing**	Multimodal LLM synthesizes telemetry into municipal title, natural language summary, and SOP plan.	Structured Work Order JSON
Stage 6	**Authority Routing Engine**	Spatial query resolves: Zone → Ward → Department → Supervisor → BVG Sanitation Team.	Assigned Team & SLA Deadline
Stage 7	**Field Dispatch**	Automated incident creation in database, push notification to field beat leader's handheld terminal.	Active Ticket (DISPATCHED)
Stage 8	**Cleaning Verification**	Post-cleaning camera re-capture or worker handheld photo comparison. AI confirms clean state.	Ticket Resolved & Audit Logged

3. Campus Surveillance Registry: Institutional Hardware Specs

Active CCTV units configured in the CSJMU production registry, reflecting standard institutional 2 MP and 1.3 MP deployments:

Camera ID	Location & Campus Zone	Resolution & FPS	Stream Protocol	Channel	Target Tensor
CAM-UIET-01	UIET Block-3 Canteen Concourse	1080p (1920×1080) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-LIB-02	Central Library Plaza Entrance	1080p (1920×1080) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-WS-03	Central Workshop North Drain	720p (1280×720) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-HST-04	Ramanujan Hostel Mess Quad	1080p (1920×1080) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-STAD-05	Stadium Pavilion South Bleachers	720p (1280×720) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-ADMIN-06	VC Secretariat VIP Porch	1080p (1920×1080) @ 15 fps	HTTP_SNAPSHOT_ISAPI	Ch 101	640×640 px

4. Network Bandwidth & Ingestion Economics

Quantitative comparison of network traffic across alternative ingestion topologies:

Ingestion Strategy	Bitrate / Size Per Cam	Daily Volume (20 Cams)	Switch Impact & Risk Profile
Continuous 4K Streams (3840×2160 @ 30fps H.264)	~12.0 Mbps per stream	**~2,592 GB / day** (240 Mbps continuous)	**Severe Congestion:** Saturates campus switches, drops VoIP/ERP traffic, requires heavy GPU decoding servers.
Continuous 1080p Streams (1920×1080 @ 15fps H.264)	~1.8 Mbps per stream	**~388 GB / day** (36 Mbps continuous)	**Moderate Overhead:** Substantially lower, but still wastes bandwidth streaming during unoccupied hours.
CSJMU Ephemeral Snapshots (Single 1080p JPEG @ 4 sweeps/day)	~180 KB per capture query	**~14.4 MB / day** (0.00017 Mbps avg)	**Optimal Institutional Fit: 99.99% bandwidth savings.** Zero network contention. Sub-second direct tensor ingestion.

5. Stratified Dynamic Random Auditing Scheduler

To prevent facilities crews from anticipating audits, CSJMU divides the active campus day (06:00 to 22:00) into four non-overlapping 4-hour operational quarters with boundary buffers:

Quarter	Shift Window	Audit Focus & Critical Targets	Trigger Offset Algorithm
1st Sweep	06:00 – 10:00	Overnight waste accumulation, hostel mess dumping, pre-class concourses.	$t \in [06:25, 09:35]$ (Gaussian offset)
2nd Sweep	10:00 – 14:00	UIET canteen peak, lunch kiosk overflow, open lawn takeaway wrappers.	$t \in [10:25, 13:35]$ (≥90 min from Sweep 1)
3rd Sweep	14:00 – 18:00	Post-afternoon dispersal, sports bleachers, library walkway bins.	$t \in [14:25, 17:35]$ (≥90 min from Sweep 2)
4th Sweep	18:00 – 22:00	Hostel dinner mess exits, evening study walkways, gate bins before night lock.	$t \in [18:25, 21:35]$ (≥90 min from Sweep 3)

6. AI Detection Classes & Volumetric Density Model

The model is fine-tuned on municipal waste datasets (TACO / ZeroWaste) to recognize 6 operational states. Each class maps to bulk density coefficients to convert visual volume into estimated kilograms:

Detection Class	Priority	Density	SLA	Standard Operating Dispatch Action
blocked_drain	CRITICAL	160 kg/m³	**30 min**	Deploy desilting squad with suction pumps and rods to avert campus storm flooding.
overflow	HIGH	120 kg/m³	**45 min**	Dispatch 2-person BVG crew with 240L wheelbarrow. Clear overflow and disinfect perimeter.
litter_pile	HIGH	90 kg/m³	**60 min**	Dispatch manual sweeping squad with litter pickers and heavy-duty polythene bags.
bin_full	MEDIUM	110 kg/m³	**90 min**	Schedule compactor truck beat round to empty bin before overflow threshold.
bin_partial	LOW	100 kg/m³	Next Shift	Logged in daily analytics. No emergency dispatch required.
bin_empty	NORMAL	N/A	Audit Pass	Recorded as positive cleanliness compliance for zone supervisor scorecard.

7. Municipal Authority Hierarchy & BVG Workforce Dispatch

Detections trigger legally enforceable municipal work orders mapped to specific ward supervisors and contractors:

Campus Precinct	Ward Designation	Departmental Owner	Beat Supervisor	Assigned Field Crew
UIET Academic Block	Ward 4 – Academic Division	Estate Office / Academic Maintenance	Er. Mahendra Gupta *(Sanitation Inspector)*	BVG Beat Squad Alpha (2 Sweep + 1 Cart)
Central Library & Admin	Ward 2 – Central Secretariat	Registrar's Office Facilities Cell	Mr. R. K. Srivastava *(Assistant Registrar)*	BVG Admin Flying Squad (VIP Beat)
Mechanical Workshops	Ward 5 – Industrial & Utilities	Works & Infrastructure Dept	Er. Suresh Verma *(Works Supervisor)*	Utility & Drain Clear Crew (Heavy Rodding)
Ramanujan Hostels	Ward 7 – Student Residences	Chief Warden Office	Dr. Pradeep Mishra *(Hostel Sanitation Supv)*	BVG Hostel Sanitation Beat (Mess Clearing)

8. Micro-Litter Detection (SAHI) & De-Duplication Logic

Micro-Litter Handling: Small cups and wrappers cover <20×20 px in wide camera frames. The system handles this through (1) prioritizing macro piles/overflow, (2) static receptacle ROI cropping (focusing on a 5m radius around bins), and (3) **SAHI (Slicing Aided Hyper Inference)** which tiles 1080p frames into overlapping 640×640 patches.
De-Duplication Gate: To prevent duplicate ticket spamming across successive sweeps, tickets are gated: should_dispatch = (confidence ≥ 0.70) AND (class != clean) AND (no_active_ticket_in_zone).
Verification: Resolved tickets require an 'After' photo from the field beat, which is verified by AI before ticket closure.

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